Overview of vector random (stochastic) processes

- A stochastic process or random process is a family of random vectors indexed by a parameter set (“time” in our case).
- For example, we might refer to a random process X_k for generic k.
  - Value of random process at any specific time $k = m$ is a random variable X_m.
- Usually assume stationarity.
  - The statistics (*i.e.,* pdf) of the RV are time-shift invariant.
  - Therefore, $\mathbb{E}[X_k] = \bar{x}$ for all k and $\mathbb{E}[X_{k_1} X_{k_2}^T] = R_X(k_1 - k_2)$.

Properties and important points

1. Autocorrelation: $R_X(k_1, k_2) = \mathbb{E}[X_{k_1} X_{k_2}^T]$. If stationary,

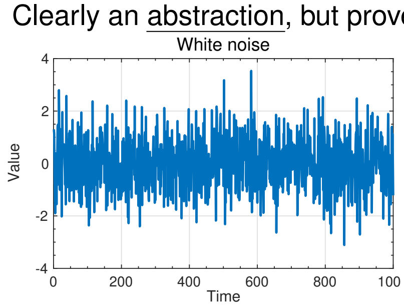
$$R_X(\tau) = \mathbb{E}[X_k X_{k+\tau}^T].$$

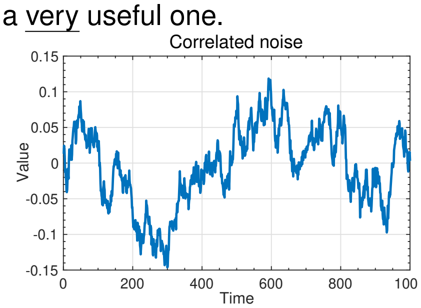
   - Provides a measure of correlation between elements of the process having time displacement τ.
   - $R_X(0) = \sigma_X^2$ for zero-mean X.
   - $R_X(0)$ is always the maximum value of $R_X(\tau)$.
2. Autocovariance: $C_X(k_1, k_2) = \mathbb{E}[(X_{k_1} - \mathbb{E}[X_{k_1}])(X_{k_2} - \mathbb{E}[X_{k_2}])^T]$. If stationary,

$$C_X(\tau) = \mathbb{E}[(X_k - \bar{x})(X_{k+\tau} - \bar{x})^T].$$

White noise

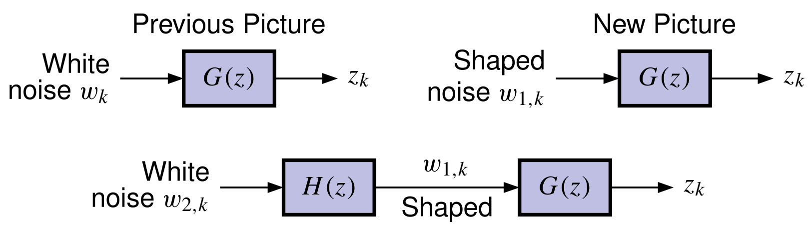
- White noise: Some processes have a unique autocorrelation:
  1. Zero mean.
  2. $R_X(\tau) = \mathbb{E}[X_k X_{k+\tau}^T] = S_X \delta(\tau)$ where $\delta(\tau)$ is the Dirac delta. $\delta(\tau) = 0 \; \forall \; \tau \neq 0$.
  - Therefore, the process is uncorrelated in time.
  - Clearly an abstraction, but proves to be a very useful one.





Shaping filters: Idea

- Will assume noise inputs to dynamic systems are white.
  - Limiting assumption, but one that can be easily fixed
    - ➠ Can use second linear system to "shape" the noise as desired.



  - Can drive our linear system with noise that has desired characteristics by introducing shaping filter $H(z)$ that itself is driven by white noise.

Shaping filters: Model

- Combined system $GH(z)$ looks exactly the same as before, *but* $G(z)$ is not driven by pure white noise any more.
  - Analysis *augments* original system model with filter states. Original system has:

$$\begin{aligned} x_{k+1} &= Ax_k + B_w w_{1,k} \\ z_k &= Cx_k. \end{aligned}$$

  - Shaping filter with white input and desired output statistics has:

$$\begin{aligned} x_{s,k+1} &= A_s x_{s,k} + B_s w_{2,k} \\ w_{1,k} &= C_s x_{s,k}. \end{aligned}$$

  - Combine into larger-order augmented system driven by white noise:

$$\begin{bmatrix} x_{k+1} \\ x_{s,k+1} \end{bmatrix} = \begin{bmatrix} A & B_w C_s \\ 0 & A_s \end{bmatrix} \begin{bmatrix} x_k \\ x_{s,k} \end{bmatrix} + \begin{bmatrix} 0 \\ B_s \end{bmatrix} w_{2,k}$$

$$z_k = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x_k \\ x_{s,k} \end{bmatrix}$$

Gaussian processes

- We will work with Gaussian noises to a large extent.
  - Uniquely defined by the first- and second central moments of the statistics ➠ Gaussian assumption not essential.
  - Our filters will always track only the first two moments.

NOTATION: Until now, we have always used capital letters for random variables.

- The state of a system driven by random process is an RV, so we could call it X_k.
- It is more common to retain standard notation x_k and understand from the context that we are discussing an RV.

Summary

- Random process is a family of RVs indexed by time.
- Will assume our random processes are stationary.
- Autocorrelation and autocovariance measure self-predictability of a signal at different time offsets.
- White noise is zero mean signal, completely uncorrelated with self ("completely random").
  - An abstraction, but a very useful one.
- If noises in a system of interest are not white, we can model them as filtered white noise to immitate the same general characteristics.
- From now on, unless stated otherwise, all noise signals will be white and Gaussian.